

Antigen processing, MHC, and lymphocyte activation

Adaptive recognition and HLA

- Clonal receptors from somatic rearrangement
- T cells see presented peptide, B cells native antigen
- Costimulation and cytokines supply context
- HLA encoded on chromosome six
- Polymorphism aids population pathogen recognition
- Co-dominant expression of inherited alleles

Class one: HLA-A, B, C

Class two: HLA-DP, DQ, DR

Complicates transplantation and disease links

MHC class one pathway

- Heavy chain, beta-two microglobulin, peptide
- Nearly all nucleated cells and platelets
- Closed groove holds short peptides
- CD8 T cells survey for intracellular change
- Proteasome, immunoproteasome in inflammation
- TAP carries peptides into the ER
- Chaperones load, edit, trim
- Stable complexes travel via Golgi to surface
- Defects impair antiviral defence

MHC class two pathway

- Alpha and beta chains
- Professional presenting cells, inducible elsewhere
- Open groove binds longer peptides
- Invariant chain blocks groove, routes to endosomes
- Autophagy delivers intracellular antigen
- B cells present peptides of captured antigen
- CD4 T cells coordinate responses

Degraded to CLIP fragment

HLA-DM removes CLIP and edits loading

HLA-DO modulates in selected cells

Broad activation, HLA, and context

- Superantigens crosslink outside peptide groove
- HLA associations neither necessary nor sufficient
- Dose, persistence, route shape fate
- Adjuvants activate sensors, not specificity
- Specificity, danger, programme, destination, duration

Many clones, massive cytokine release

Typing guides transplantation, drug safety

Trafficking and T-cell fates

- Enter nodes via high endothelial venules
- Unactivated cells exit on S1P gradients
- Activation retains, then changes homing
- Type 1: interferon gamma, intracellular pathogens
- Type 2: interleukins 4, 5, 13, helminths
- Type 17: interleukins 17, 22, barriers
- Follicular helpers give CD40 ligand
- FOXP3 regulatory cells suppress
- Categories flexible, mixed states occur
- CD8 kill via perforin, granzyme, death receptors

Induce interleukin 2 and high-affinity receptor

Drugs target calcineurin, mTOR, nucleotides

B-cell activation

- Membrane immunoglobulin binds native antigen
- CD19 co-receptor amplifies complement-tagged signal
- T-dependent: present peptide to follicular helpers
- T-independent: rapid but little memory

Extrafollicular: plasmablasts and IgM

Germinal centres: affinity, switching, memory

Especially weak in young children

Protein conjugation recruits T-cell help

Expansion, memory, and synapse

- Clonal expansion then contraction
- Memory alters threshold, location, metabolism
- Memory need not be sterilising
- Synapse: LFA1-ICAM stabilises contact
- Granules polarised to limit bystander damage

Evasion and cross-presentation

- Viruses block proteasome, TAP, loading, surface
- Low class one removes natural killer inhibition
- Dendritic cells present external antigen on class one

Evolutionary tension

Primes killers against non-infecting viruses, tumours

T-cell activation

- Alpha-beta receptor with CD3 chains
- Recognises combined peptide-MHC surface
- CD4 binds class two, CD8 class one
- Modest affinity, sensitive via microclusters
- CD28 binds B7 as costimulation
- CTLA4 and PD1 restrain activation

Without it: anergy, deletion, tolerance